symmetric in the indices a and b. This suggests that these particular solutions cannot capture the full content of the Wheeler-DeWitt equation.

2. The operator products in (4.5.21-23) are ill-defined and contain factors of $\delta^3(0)$. The problem of how to regularise these operators is a difficult one and is currently the subject of some debate. For example, to define a point-split operator it is necessary to introduce a background metric and curve, and the ultimate effect of this unwanted background structure is not clear.
3. The explicit dependence of the solutions on loops in Σ means they are not Diff(Σ) invariant. It is ironic that, in this formalism, the Wheeler-DeWitt equation can be solved fairly easily but the Diff(Σ) constraints—which are regarded as being relatively innocuous in the conventional approach—should now appear so intractable!

4.5.3 The Rovelli-Smolin Variables

One of the most important recent advances in the use of the Ashtekar variables is the development by Rovelli and Smolin (1988) of a new canonical algebra for general relativity. The basic objects are gauge-invariant functions defined on the phase space of the system by

$$T_\eta[\tilde{e}, A] := \mathrm{Tr}\Big(\mathcal{P}\exp\oint_\eta A\Big) \tag{4.5.26}$$

$$T^a_{\eta,s}[\tilde{e}, A] := \mathrm{Tr}\Big(\mathcal{P}\tilde{e}^a(\eta(s))\exp\oint_\eta A\Big) \tag{4.5.27}$$

where $\eta : S^1 \to \Sigma$ is a smooth loop in Σ and $s \in S^1$ is a particular parameter value. There are analogues for any number of points on the loop. For example, the function with two insertions is

$$T^{a_1 a_1}_{\eta;s_1,s_2}[\tilde{e}, A] := \mathrm{Tr}\big(\tilde{e}^{a_1}(\eta(s_1))\mathcal{P}_\eta(s_1, s_2)\tilde{e}^{a_2}(\eta(s_2))\mathcal{P}_\eta(s_2, s_1)\big) \tag{4.5.28}$$

where $\mathcal{P}_\eta(s_1, s_2)$ is the parallel transport operator from the point $\eta(s_1)$ to the point $\eta(s_2)$ along the loop η.

The first important property of these "T-variables" is that they generate *all* functions on the phase space. The second, and crucial, property is that they have a closed Poisson bracket algebra which can be written somewhat symbolically as

$$\{T^n, T^m\} \sim T^{n+m-1} \tag{4.5.29}$$

where the superscript represents the number of insertions (thus (4.5.26), (4.5.27) and (4.5.28) are T^0, T^1 and T^2 functions respectively). Note that the T^0 and T^1 variables generate a closed subalgebra of the complete T-algebra.

The central idea of the Rovelli-Smolin programme is to use this algebra as the basic algebra whose representations determine the quantum theory. In particular, they construct a type of Fock space quantisation in which the analogue of a "n-particle" state is a function $\psi(\eta_1, \eta_2 \ldots \eta_n)$ of n loops. The T^0 variable acts like a creation operator, for example

$$(T_\eta \psi_2)(\xi) := \psi_2(\eta, \xi), \tag{4.5.30}$$

whereas the T^1 operators map each n-loop sector into itself. By this means Rovelli and Smolin succeed in constructing a deformation (with powers of $\hbar$ as coefficients) of the classical T-algebra.

The next major step is to construct the constraints (4.5.13-14) as a limit of sequences of these T-variables. Rovelli and Smolin then show that the longitudinal constraints $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$ (*i.e.*, the Wheeler-DeWitt equation) can be satisfied provided the Fock-space functions $\psi(\eta_1, \eta_2 \ldots \eta_n)$ are concentrated on smooth, non-intersecting loops.

Of course, these states are not Diff(Σ) invariant because the diffeomorphism group moves the loops around. However, a state that *is* Diff(Σ)-invariant can be found by requiring the n-loop functions $\psi(\eta_1, \eta_2 \ldots \eta_n)$ to be constant on the Diff(Σ) orbits, which are the *link* classes of the manifold Σ. Thus, by these means, Rovelli and Smolin claim to have constructed exact solutions to *all* the constraints of quantum gravity.

This is a most impressive result but, as always, there are various difficulties tucked away in the formalism.

1. The Fock space construction is rather formal. In particular, it is not known what inner product should be placed on the states.
2. The classical T-variables are not real and, correspondingly, there is no reason to expect the quantum operators to be hermitian. This has several drawbacks:
   a) The problem of deciding what inner product to choose is made much harder.
   b) It is difficult to construct any general representation theory of the T-algebra. This means it is also hard to assess the significance of the particular Fock-like representation that has been found.

3. The constraints are singular limits of the T variables and generate the usual unpleasant problems of operator-ordering and regularisation.
4. It is difficult to relate the T-variables to anything in conventional general relativity. This makes it difficult to interpret the quantum results.

Much more could be said about this approach and the best ways of tackling the problems which arise. The latter are certainly non-trivial but, nevertheless, the Rovelli-Smolin algebra is one of the most intriguing ways yet suggested of solving the constraints of canonical quantum gravity.

5. The Problem of Time in Quantum Gravity

5.1 General Remarks

5.1.1 The Major Issues

The question of 'time' is one of the deepest issues that must be faced in a quantum theory of gravity. It is intimately related to a whole variety of interpretational problems, including

1. the status of the notion of *probability* and the extent to which it can be said to be *conserved*;
2. the associated concepts of *causality* and *unitarity*;
3. the extent to which *spacetime* is a meaningful concept;
4. the way in which our "classical" world *emerged* from some primordial quantum event (in a theory of quantum cosmology).

These difficulties stem from the variety of roles which time plays in the technical and conceptual frameworks of conventional physics:

1. Time is "Heraclitean" in the sense that "time is that which allows contradictory things to occur" (Unruh and Wald 1989). For example, a particle cannot be at different positions at the same time, but it can be at the same position at different times.
2. Time is the parameter with respect to which change is measured.
3. In the case of quantum theory we have in addition that

a) the scalar product of states is defined at a single time;

b) complete sets of commuting observables are defined at a single time;

c) measurements are carried out at a single time, and states specify the probabilities of the results of such measurements.

Several important questions arise concerning these properties of time in the case of a quantum theory of gravity.

1. To what extent can time continue to play these roles?
2. In particular, is the concept of time *exact*, or can it only be implemented in some *approximate*, perhaps semi-classical, sense? If so, the notions of probability, unitarity, causality and spacetime will also only be approximate.

   But if 'probability' is not an exact concept does this not mean that quantum theory itself is only approximate? If so, all existing approaches to quantum gravity seem doomed from the outset!
3. In classical general relativity, time is only a local coordinate, and a manifold may need to be covered by more than one coordinate chart. How, if at all, is this reflected in the quantum theory of gravity?

5.1.2 The Role of Background Structure

In conventional quantum physics, 'time' is not an observable in the normal sense since it is not represented by any operator but is instead treated as a background parameter which, as in classical physics, is used to mark the evolution of the system. This is true not only of non-relativistic quantum theory but also of relativistic particle dynamics and quantum field theory. Thus 'time' is arguably part of the classical background which plays such a crucial role in the "Copenhagen" interpretation of the theory.

It is not surprising therefore that, within any particular approach to quantum gravity, the nature of the problem of time depends strongly on the type of background structure that is present. For example, there is no particular difficulty in the particle physics schemes since time can be measured with respect to the background Minkowski metric, thus taking us back to the familiar realm of special relativity and quantum field theory.

Of course, whether or not such a measure of time is *physically* correct is another matter. This is part of the general question of the extent to which the spacetime

concepts of classical general relativity can be properly captured by a weak-field perturbation around a Minkowskian background. For example, the peculiar behaviour of the lightcones (and therefore time) at the event horizon of a black hole is not the easiest thing to reproduce in a graviton-based picture.

Similar remarks apply to extensions of the particle physics approach in which the background is a more general spacetime manifold $\mathcal{M}$ with the gravitational fields—real or virtual—being required to be asymptotically flat. In such a situation it may be possible to define an asymptotic time which can be used to discuss a graviton S-matrix but, once again, it is not clear that this limited notion of time is sufficient to address all the questions we might wish to raise in a quantum theory of gravity.

In the canonical approach to quantum gravity the question of 'time' is very different. The background is now the *three*-dimensional manifold Σ. If Σ is non-compact then, as above, the asymptotic structure might be used to define an absolute time that is linked to the surface terms ignored in the discussions in §4.

However, in the "cosmological" case in which Σ is compact, the technical heart of the theory is the constraints on the $12\infty^3$ canonical variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$ defined on Σ, and the notion of time (and therefore spacetime) has to be *extracted* in some way from these variables, or from any matter fields or particles that may be added to the system. This is the question with which we will be concerned in the rest of this chapter.

The way in which time is related to the canonical variables depends rather on the approach to canonical quantisation that is being adopted.

1. In the scheme in which the system is reduced to true canonical form before it is quantised (§4.4.1), the definition of time is linked to the imposition of a gauge. This can take the form of
   a) an explicit definition of time as some functional of the canonical variables;
   b) a more implicit relation in which a gauge is chosen by requiring some function of the canonical variables to vanish, or
   c) by specifying the lapse function (or its time derivative) to be some specific functional of the canonical variables.
2. In the alternative approach, all $12\infty^3$ variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$ are quantised and the constraints are imposed as restrictions on the state vectors (§4.4.2). This leads

to the Wheeler-DeWitt equation whose interpretation (which we have studiously ignored so far) is deeply related to the meaning of time. Two conflicting views arise at this point:

a) It is possible to extract a concept of time from the Wheeler-DeWitt equation that is exact and which applies at all levels of the theory.

b) The notion of time is only valid in some semi-classical sense which can be extracted from WKB solutions to the Wheeler-DeWitt equation.

5.2 Time as a Classical Variable

5.2.1 The Main Idea

The idea is to employ the approach to canonical quantum gravity discussed in §4.4.1 in which the theory is reduced to a true canonical form before it is quantised. This involves gauge-fixing which, in the present context, means defining a time variable in terms of the gravitational or matter content of the theory.

For example, in a cosmological model, one might like to define time as the functional

$$T[\gamma] := \int_{\Sigma} \left(\det \gamma(x)\right)^{\frac{1}{2}} d^3x \tag{5.2.1}$$

which says that

1. the manifold Σ is always a surface of equal time;
2. the value of this time is equal to the volume of the universe.

Thus, rather than referring to, for example, the simultaneous value of a scalar field ϕ and the volume of the universe at some background time t, one speaks instead of the value of ϕ when the volume of the universe takes on various values.

Such a choice works in a minisuperspace model in which the Diff(Σ) action has already been gauge-fixed (*e.g.*, by choosing $N^a = 0$), but in general we need to remove $1\infty^3$ variables, which involves dropping the requirement that Σ is automatically a hypersurface of equal time. For example, one could define [23]

$$T(x;\gamma] := \left(\det \gamma(x)\right)^{\frac{1}{2}} \tag{5.2.2}$$

[23] The notation $T(x;\gamma]$ means that T is a function of x and a functional of γ.

which says that the value of time at a point $x \in \Sigma$ is equal to $\left(\det\gamma(x)\right)^{\frac{1}{2}}$. This "multi-fingered" time (Kuchar, 1972) is analogous to the Tomonaga multi-time formulation of conventional quantum field theory.

The most general definition of a time function is a functional $T(x;\gamma,\pi]$ on the *phase space* of the system with the interpretation that the value of 'time' at the point $x \in \Sigma$ is equal to $T(x;\gamma,\pi]$. The canonical variables of matter fields can also appear as arguments of T.

5.2.25 Reduction to Canonical Form

The key step is to reduce the constrained system to canonical form. In the Hamiltonian framework we are adopting this involves the following steps.

1. Find the momentum π_T which is conjugate to T so that

$$\{T(x), \pi_T(x')\} = \delta(x, x') \tag{5.2.3}$$

   for each point $x \in \Sigma$.

2. At each $x \in \Sigma$, solve the classical constraint $\mathcal{C}_\perp(x) = 0$ for π_T as a functional of T and the canonical variables. This functional becomes the Hamiltonian density $H_T(x;\gamma,\pi,T]$ for a conventional canonical system. [24]

3. This system can then be quantised in a standard way and leads to the Tomonaga-Schrödinger equation

$$i\hbar\frac{\delta\Psi}{\delta T(x)} = \widehat{H}_T(x)\Psi \tag{5.2.4}$$

One of the real advantages of this approach is the relative absence of any ambiguity about what the final quantum theory is supposed to mean. The state-vectors give the probability distributions of the results of measurements on the variables that remain after the "gauge" variables have been removed. In particular, there should be no difficulty in imposing a genuine Hilbert space structure on the quantum states.

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[24] Strictly speaking, it is first necessary to handle the Diff(Σ) constraints $\mathcal{C}(\xi) = 0$. This can be done by choosing three functionals $S^a(x;\gamma,\pi]$ to serve as coordinates on Σ, or a hybrid scheme can be used in which the spatial constraints are imposed in the form $\widehat{\mathcal{C}}(\xi)\Psi = 0$.

5.2.3 A Minisuperspace Model

Unfortunately, a number of problems arise with this scheme which can be illustrated with the aid of the simple superspace model introduced at the end of §4 (Blyth and Isham 1975). We will choose the cosmological case $k = 1$ where Σ is a three-sphere. Note that there is no explicit x-dependence in this model since, in effect, we have already chosen the Diff(Σ) gauge $N^a(x) = 0$. Thus the question of a multi-fingered time does not arise.

One natural choice for a global "cosmological" time is the radius a of the universe. Thus, according to the discussion above, the reduced Hamiltonian for this system can be found by solving the constraint

$$\frac{-\pi_a^2}{24a} - 6a + \frac{\pi_\phi^2}{2a^3} + a^3 V(\phi) = 0 \tag{5.2.5}$$

for the variable π_a. Hence

$$H_a = \pm\sqrt{24}\big(-6t^2 + \frac{\pi_\phi^2}{2t^2} + t^4 V(\phi)\big)^{\frac{1}{2}}. \tag{5.2.6}$$

Another possible choice is the "conjugate" variable $t := \pi_a$ which (in the simple case $V(\phi) = 0$) leads to the Hamiltonian

$$H_{\pi_a} = \pm\left(\frac{-\frac{1}{12}t^2 \pm (\frac{1}{144}t^4 + 48\pi_\phi^2)^{\frac{1}{2}}}{24}\right)^{\frac{1}{2}}, \tag{5.2.7}$$

while a natural "matter" definition is $t = \phi$ which yields

$$H_\phi = \pm\big(\frac{a^2\pi_a^2}{12} + 12a^4 - 2a^6 V(t)\big)^{\frac{1}{2}}. \tag{5.2.8}$$

A number of general problems can already been seen from these simple examples.

1. The constraint equation is typically an algebraic equation for the momentum π_T conjugate to the time variable, and therefore has more than one solution. In effect, we are factorising the classical constraint in the form $\mathcal{C}_\perp = (\pi_T - \alpha_1)\ldots(\pi_T - \alpha_r)$ and then choosing just one of the roots α_i to be the reduced Hamiltonian. This choice is partly determined by various physical requirements: for example, the reduced Hamiltonian should be a positive, hermitian operator. However, there may still be some residual ambiguity in the procedure.
2. The reduced Hamiltonian can be a very non-linear function of the canonical operators and severe operator-ordering problems occur.

3. The reduced Hamiltonian is time dependent. This has several effects.
   a) A time-dependent Hamiltonian means that energy can be fed into, or taken out of, the quantum system. In a normal physical situation this arises because the system is not closed, and then the time-dependence of the Hamiltonian is determined by the environment. However, a compact three-manifold Σ (the "universe") has no external environment, and so the time-dependence is perhaps a little odd.
   b) If $H(t)$ is time-dependent, the Schrödinger equation

$$i\hbar\frac{d\psi_t}{dt} = \widehat{H}(t)\psi_t \tag{5.2.9}$$

   does not lead to the simple second order equation

$$-\hbar^2\frac{d^2\psi_t}{dt^2} = (\widehat{H}(t))^2\psi_t \tag{5.2.10}$$

   since there is an extra term involving $\frac{d}{dt}\widehat{H}(t)$. In particular, this means that the Schrödinger equations above do *not* imply the Wheeler-DeWitt equation associated with the constraint (5.2.5). This is not necessarily a bad thing but it does show the inequivalence of the different approaches to the canonical quantisation of gravity.
3. There is no natural choice for the time variable and no reason to suppose that different choices will give equivalent quantum theories. This contrasts strongly with the situation in conventional quantum field theory where different global Lorentz frames lead to unitarily equivalent canonical quantum theories.
4. The square roots can be handled as operators if the object whose square root is being taken is hermitian and positive. However, in general this will only be true for a limited range of t (which depends on $V(\phi)$). It might still be possible to take the square-root outside this range but the result will certainly not be hermitian, resulting in a time evolution operator that is non-unitary.

This last problem has been analysed in depth in a series of papers by Hajicek (1986, 1988, 1989, 1990a, 1990b) which explore the general question of the construction of time functions on the phase space of a constrained system including, for example, the existence of solutions to (5.2.3).

The failure of time functions to be global seems to be a generic property of systems of this type but it is not necessarily a disaster. Much work has been done in recent

years on quantum theories of the creation of the universe and, in any such theory, something peculiar must necessarily happen to time near the origination event. For example, in the Hartle-Hawking (1983) and Vilenkin schemes (1988) there is a sense in which time becomes imaginary, and the non-unitarity of the canonical picture may sometimes be an analogue of this effect.

5.3 Time and the Wheeler-DeWitt Equation

5.3.1 Internal Time and the Meaning of $\Psi[\gamma]$

For the reasons above most of the discussion of time in canonical quantum gravity has been within the framework in which all $12\infty^3$ variables $(\gamma_{ab}(x), \pi^{cd}(x))$ are quantised and with the constraints imposed on the state vectors as in (4.4.12-13). In the representation on functionals $\Psi[\gamma]$ this leads to the Wheeler-DeWitt equation , and the problem is to understand the relation of this equation to the concept of time.

An appropriate starting point is to consider the physical meaning of a state-functional Ψ. The operators $\widehat{\gamma}_{ab}(x)$ and $\widehat{\pi}^{cd}(x)$ are defined as in (4.4.16–17) in order that they be *hermitian* with respect to the inner product defined using the "measure" [25] $\mathcal{D}[\gamma]$ on the space $\mathrm{Riem}(\Sigma)$ of all Riemannian metrics on Σ (for our present purposes, the technical problems mentioned in §4.4.5 that are associated with this assumption can be ignored). Similarly, in discussing whether or not the constraint operators $\widehat{\mathcal{C}}_{\perp}(x)$ and $\widehat{\mathcal{C}}_{a}(x)$ are, or should be, hermitian, it is the inner product

$$\langle \Psi_1 | \Psi_2 \rangle := \int_{\mathrm{Riem}(\Sigma)} \Psi_1^*[\gamma] \Psi_2[\gamma] \, \mathcal{D}[\gamma] \tag{5.3.1}$$

on $L^2\big(\mathrm{Riem}(\Sigma), \mathcal{D}[\gamma]\big)$ which is assumed.

In analogy with normal canonical quantum field theory, this suggests an interpretation of Ψ in which, if a measurement is made of the metric-tensor on the three-manifold Σ, [26] the probability of finding the result in a measureable subset B of $\mathrm{Riem}(\Sigma)$ is

$$\mathrm{Prob}(\gamma \in B; \Psi) := \int_B \big|\Psi[\gamma]\big|^2 \, \mathcal{D}[\gamma]. \tag{5.3.2}$$

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[25] The notation $\mathcal{D}[\gamma]$ refers to the non-existing Lebesgue measure on $\mathrm{Riem}(\Sigma)$. I will keep the symbol $d\mu[\gamma]$ for a genuine measure.

[26] Of course, it is not really feasible to measure the gravitational field *everywhere* on Σ, since that is the entire universe! However, (5.3.2) is still meaningful if a trace is taken over the fields on the regions of Σ where no measurements are made. This produces an effective density matrix for the state in the restricted region where measurements *are* made.

If a matter field ϕ is present then the state vector is of the form $\Psi[\gamma, \phi]$, with an analogous interpretation.

This view of $|\Psi[\gamma]|^2$ as a probability density for three-metrics may seem plausible and, indeed, an interpretation of this type has been used a number of times, especially in the context of quantum cosmology. However, as mentioned in §4.4.5, the counting of degrees of freedom is wrong.

A metric tensor γ_{ab} corresponds to $6\infty^3$ degrees of freedom, *i.e.,* six degrees of freedom per spatial point in Σ. Three of these are associated with the action of Diff(Σ) and can be removed by an explicit gauge-fixing or by imposing the constraints $(\widehat{\mathcal{C}}_a(\xi)\Psi)[\gamma] = 0$ as in (4.4.18). In the latter case, the state vector is defined on superspace Riem(Σ)/Diff(Σ), which we will denote by writing $\Psi({}^{(3)}\mathcal{G})$. The inner product (5.3.1) must be replaced by a measure on Riem(Σ)/Diff(Σ) which, following the example of Yang-Mills theory, can be done in a variety of ways. The state-functional $\Psi({}^{(3)}\mathcal{G})$ then has the interpretation of a probability amplitude for three-*geometries*,*i.e.,* equivalence classes of metric tensors with respect to the action of Diff(Σ).

However, the counting is still not correct since a three-geometry ${}^{(3)}\mathcal{G}$ has $6\infty^3 - 3\infty^3 = 3\infty^3$ degrees of freedom whereas the true gravitational system should have a count of $2\infty^3$. The reason for this discrepency can be seen from two perspectives:

1. We still have to impose the constraints $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$, *i.e.,* solve the Wheeler-DeWitt equation. This will remove a further $1\infty^3$ degrees of freedom, leaving $3\infty^3 - 1\infty^3 = 2\infty^3$ as required.
2. From our earlier discussion, we know that the concept of 'time' has to be extracted from the system in some way. In the present context this means that $1\infty^3$ of the $3\infty^3$ degrees of freedom are associated with a multi-fingered time $T(x;\gamma]$ which, once identified, will leave the correct count of $2\infty^3$.

The close relation between these two positions can been seen clearly with the aid of our minisuperspace model. The first view says we must impose the constraint leading to the Wheeler-DeWitt equation (4.4.40). However, the second view says that some (in this case, one) of the configuration variables should be identified with 'time'. For example, if Ω is chosen for this purpose (we recall that $\Omega := \log a$ where a is the radius of the three-sphere) then the Wheeler-DeWitt equation (4.4.40) is to be interpreted